QUANT

Mensuration & Geometry

(Triangles, Polygons & Mensuration)

Assignment 05

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Geometry Online Assignment Cluster I -05

1. 3 cubes having an edge of 7 cm are placed together, thus resulting into a cuboid. Find the volume of the cuboid (in cc).

1. 686 2. 294 3. 1129 4. 1029

2. What is the volume of a largest possible cylinder, if it is shaped by rolling a square piece of area 484sq cm length wise?

1. 647 cc 2. 847 cc 3. 947 cc 4. 747 cc

3. The surface area of a cubical room is numerically equal to its volume. A rod is having a length of 10. Is it possible to keep that rod inside that room?

1. Yes 2. No
3. Depends upon the height of the cube 4. Depends upon the material of the rod

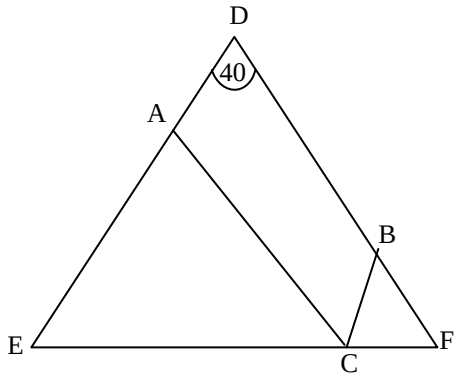
4. Instead of walking along two adjacent sides of a rectangular field, a boy took a short cut along the diagonal and saved a distance equal to half the longer side. Then the ratio of the shorter side to the longer side is

1. 1 : 2 2. 2 : 3 3. 1 : 4 4. 3 : 4

5. In the figure (not drawn to scale) given below, if AD = CD = BC, and $\angle BCE = 96^o$, how much is $\angle DBC$?

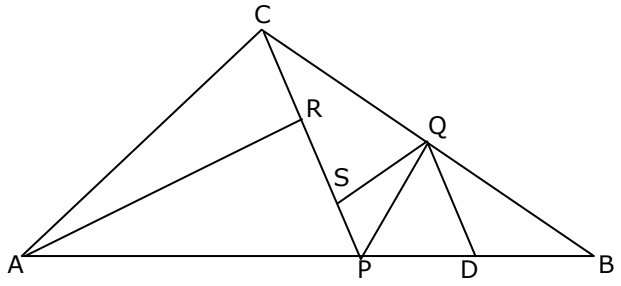
1. 32° 2. 84° 3. 64° 4. Cannot be determined

6. In the triangle DEF below, EC = AC, CF = BC. Angle D = 40°. What is the measure of angle ACB?



1. 140° 2. 70° 3. 100° 4. None of these

7. In the figure (not drawn to scale) given below, P is a point on AB such that AP: PB = 4 : 3. PQ is parallel to AC and QD is parallel to CP. In Δ ARC, ∠ARC = 90°, and in ΔPQS, ∠PSQ = 90°. The length of QS is 6 cm. What is ratio AP : PD?



1. 10 : 3 2. 2 : 1 3. 7 : 3 4. 8 : 3

8. How many complete balls of surface area 196π sq. cm each can be just packed in a hollow cylindrical open container of internal diameter 14 cm and internal height 84.05 cm?

1. 4 2. 5 3. 6 4. 7

9. Δ ABC is a right angled triangle with ∠ B = 90^0, AB = 20 cm and BC = 21cm. A circle with centre O is inscribed in Δ ABC. OD, OE, OF are perpendiculars drawn on the sides AB, BC and CA respectively. Find the ratio of the area of the quadrilateral FOEC to the area of quadrilateral ADOF.

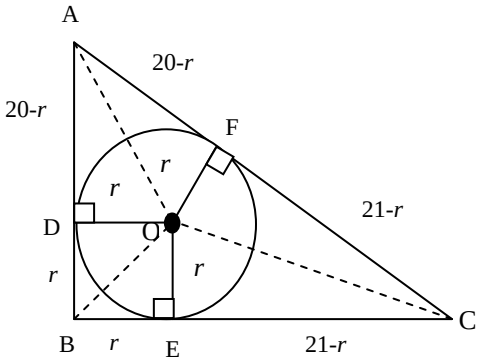
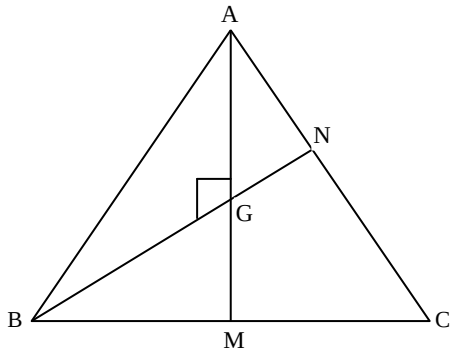
1. 15 : 14 2. 14 : 15 3. 12 : 11 4. 7 : 5

10. In a Δ ABC, two medians, AM of length 13 cm and BN of length 12cm intersect at point G. If the Δ AGB so formed is a right angled triangle with AB as the hypotenuse, then find the area of Δ ABC.

1. 120 cm^2 2. 100 cm^2 3. 104 cm^2 4. 106 cm^2

Answer key & Explanations

Q. No.	Ans.	Exp.
1.	4	The length of the cuboid will be 21cm.

		Its height and width will remain same as that of each of the original cubes i.e. 7 cm. Now you can find the volume. So volume = 21 × 7 × 7 = 1029 cu cm.
2.	2	The side of the square is 22 cm. When it is converted into a cylinder by rolling length wise, the length of 22 will become the base circumference of the cylinder and the breadth of the square will become the height of the cylinder. So $2\pi r = 22$ $\Rightarrow r = 7/2$. Hence the volume of the cylinder is 22/7 × 7/2 × 7/2 × 22 = 121 × 7 = 847 cu cm.
3.	1	$6a^2 = a^3 \Rightarrow a = 6$. The length of the longest rod that can be kept inside this room = $6\sqrt{3} = 6 \times 1.7 = 10.2$. This implies the rod having a length of 10 can be kept in this room,
4.	4	In such case, the best way is to solve with the help of options. Besides options just keep in mind the Pythagoras triplets also. The moment you see the fourth option, in a moment you can realize that the length of the diagonal will be 5 and 5 is 2 less than 4 + 3 = 7. You can verify that 2 is half of 4, the longer side and hence 4th option is the answer.
5.	3	Take ∠DAC as x°, ∠ACD also becomes x° (both are equal). Take ∠BDC as y°, ∠ CBD also becomes y°. ∠ DCB take it as z° and ∠ CDA becomes (180 – y)°. Now as per the exterior angle formula, x + (180 – y) = 96 + z. Secondly x + 96 + z = 180°, thirdly x + x + (180 – y) = 180°. Solve these three simple equations and get the value of y as 64°.
6.	3	Let ∠EAC & ∠CEA be y° and ∠CBF & ∠BFC = x° and ∠ACB = z°. Now in quadrilateral ADBC, (180° - x) + 40° + (180° - y) + z = 360° ⇒ x + y – 40 = z (i). Otherwise in the triangle DEF, ∠40 + ∠ y + ∠ x = 180 ⇒ x + y = 140° (ii) the value of x + y in (i) and get the value of z as 100°.
7.	3	The ratio of CQ : QB will also be 4 : 3. Now this is also equal to the ratio of PD : DB (Similar triangles BDQ and BPC). So PD is $4/7^{th}$ of PB. So ratio will be 4 : 3 × 4/7 = 7 : 3.
8.	3	Calculate the radius of the ball. So $4\pi r^2 = 196\pi \Rightarrow r = 7$, diameter = 14. Now this 14 is matching with the internal diameter of the container. The height of the container is given to be 84.05, when it is divided by 14 you get ≈ 6.
9.	1	 AC= $\sqrt{AB^2 + BC^2} = 29cm$ ∴ (20 – r) + (21 – r) = 29 ⇒ r = 6 Δ OEC & Δ OFC are congruent Area of (Quad FOEC) ≡ 2 (Area Δ OEC)

		$= 2 \times \frac{1}{2} \times 15 \times 6$ = 90 cm^2 Similiarly Δ AOD & Δ AOF are congruent ≡ Δ AOF (RHS) Area of (Quad ADOF)= 2 (AreaΔAOD) $= 2 \times \frac{1}{2} \times 14 \times 6 = 84cm^2$ $\frac{Area(QuadFOEC)}{Area(QuadADOF)} = \frac{90}{84} = \frac{15}{14}$
10.	3	Centroid G divide median in the ratio of 2 : 1 AM = 13 ⇒ AG = $\frac{2}{3}(13) = \frac{26}{3}$ cm BN = 12 cm, Area (ΔABN) = $\frac{1}{2} \times BN \times AG = \frac{1}{2} \times 12 \times \frac{26}{3}$ = 52 cm^2 Median BN divides Δ ABC into two ΔS of equal area ∴ Area (Δ ABC) = 2(52) = 104 cm^2